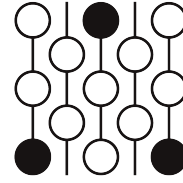


Review of Submission 3C

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

Correctness. The model arrives at the conjecturally correct range of probabilities p , but falsely claims a complete proof. The counterexamples for $p \in (1/3, 1/2)$ and $p \in (1/2, 1)$ are presented correctly. The immediate cases of $p = 0, 1$ and the short argument via symmetry for $p = 1/2$ are also noted. However, the proof of the remaining case $p \leq 1/3$ is reduced to two lemmas, neither of which follow from the provided citations, although the reduction is given correctly. Upon further inspection, the lemmas are plausible but proofs remain open to the best of my knowledge.

Novelty. The novelty of the argument is limited at best. The approach taken is rather naive, and after short manipulations the conclusion is made on the basis of two falsely cited references. (In hindsight, the main connection to the Manickam-Miklos-Singhi conjecture is not made.)

Presentation. The presentation of the paper is very bare bones. The paper includes some citations, there is no attempt at exposition, and the paper reads like a list of lemmas and algebraic manipulations. This may be due in part to the simplicity of the argument, which requires only a sequence of routine steps.

2 Recommendation

I recommend a decision of rejection. While the argument does reduce the problem to two concrete lemmas, it is not clear that the lemmas are more approachable than the original question. This combined with the lack of novelty in the reduction leads to my recommendation.

3 Submission with inline comments

Abstract

We determine all $p \in [0, 1]$ such that, for every finite probability vector $(w_1, \dots, w_m)$ and i.i.d. Bernoulli(p) random variables $v_1, \dots, v_m$, one has

$$\mathbb{P}\left(\sum_{i=1}^m w_i v_i \geq p\right) \geq p.$$

The answer is

$$p \in [0, 1/3] \cup \{1/2, 1\}.$$

External inequalities used

We use two standard sharp inequalities.

Lemma 1 (Hoeffding–Pinelis comparison for weighted Bernoulli sums). *Let $0 < \theta < 1$, let $\xi_1, \dots, \xi_m$ be i.i.d. Bernoulli(θ), and let $a_i \geq 0$, $\sum_i a_i = 1$. Then for every real x ,*

$$\mathbb{P}\left(\sum_{i=1}^m a_i \xi_i < x\right) \leq \sup_{n \geq 1} \mathbb{P}\left(\frac{1}{n} B_{n,\theta} < x\right),$$

where $B_{n,\theta} \sim \text{Bin}(n, \theta)$.

This is the Bernoulli specialization of Hoeffding’s extremal theorem for sums of bounded independent random variables; see Hoeffding [1, Theorem 4]. For the left-tail formulation in essentially the above form, see also Pinelis [3, Corollary 3.7].

Reviewer 1

The citations provided do not imply the stated lemma. [1, Theorem 4] concerns sums of Bernoulli random variables with different means, and [3, Corollary 3.7] does not even exist. If true, a proof or correct citation is needed (I was unable to find one myself).

Lemma 2 (Ramanujan–Jogdeo–Samuels binomial inequality). *If $0 \leq \theta \leq 1/3$ and $B_{n,\theta} \sim \text{Bin}(n, \theta)$, then*

$$\mathbb{P}(B_{n,\theta} < n\theta) \leq 1 - \theta$$

for every integer $n \geq 1$.

Equivalently,

$$\mathbb{P}(B_{n,\theta} \geq n\theta) \geq \theta.$$

This is a classical binomial-tail inequality proved by Jogdeo and Samuels [2]; it is often discussed as a Ramanujan-type binomial inequality.

Reviewer 1

Same as above, Lemma 2 is not immediate from the given citation. It is not clear if this statement is known in the literature either.

Main theorem

Theorem 1. *Let $p \in [0, 1]$. The following assertion holds for every $m \geq 1$, every choice of nonnegative weights $w_1, \dots, w_m$ with $\sum_i w_i = 1$, and i.i.d. Bernoulli(p) random variables $v_1, \dots, v_m$:*

$$\mathbb{P}\left(\sum_{i=1}^m w_i v_i \geq p\right) \geq p.$$

This is true if and only if

$$p \in [0, 1/3] \cup \{1/2, 1\}.$$

Proof. We prove sufficiency and necessity separately.

Sufficiency for $0 \leq p \leq 1/3$. The case $p = 0$ is trivial, since the event $\sum_i w_i v_i \geq 0$ has probability 1. Assume $0 < p \leq 1/3$, and set

$$S = \sum_{i=1}^m w_i v_i.$$

By Lemma 1, with $\theta = p$ and $x = p$,

$$\mathbb{P}(S < p) \leq \sup_{n \geq 1} \mathbb{P}(B_{n,p} < np).$$

By Lemma 2, each term in the supremum is at most $1 - p$. Therefore

$$\mathbb{P}(S < p) \leq 1 - p,$$

and hence

$$\mathbb{P}(S \geq p) = 1 - \mathbb{P}(S < p) \geq p.$$

Sufficiency for $p = 1/2$. Let

$$S = \sum_{i=1}^m w_i v_i.$$

Since $v_i \sim \text{Bernoulli}(1/2)$, the vector $(1 - v_1, \dots, 1 - v_m)$ has the same distribution as $(v_1, \dots, v_m)$. Thus $1 - S$ has the same distribution as S . Hence

$$\mathbb{P}(S > 1/2) = \mathbb{P}(S < 1/2).$$

Consequently,

$$\mathbb{P}(S \geq 1/2) = \mathbb{P}(S > 1/2) + \mathbb{P}(S = 1/2) = \frac{1 + \mathbb{P}(S = 1/2)}{2} \geq \frac{1}{2}.$$

Sufficiency for $p = 1$. If $p = 1$, then $v_i = 1$ almost surely for every i , so

$$\sum_i w_i v_i = \sum_i w_i = 1$$

almost surely. Thus

$$\mathbb{P}\left(\sum_i w_i v_i \geq 1\right) = 1.$$

Necessity: failure for $1/3 < p < 1/2$. Take $m = 3$ and $w_1 = w_2 = w_3 = 1/3$. Let

$$K = v_1 + v_2 + v_3 \sim \text{Bin}(3, p).$$

For $1/3 < p < 1/2$,

$$\frac{K}{3} \geq p \iff K \geq 2.$$

Therefore

$$\mathbb{P}\left(\frac{K}{3} \geq p\right) = \mathbb{P}(K \geq 2) = 3p^2(1-p) + p^3 = p^2(3-2p).$$

Now

$$p^2(3-2p) < p$$

is equivalent to

$$p(3-2p) < 1,$$

or

$$2p^2 - 3p + 1 > 0,$$

i.e.

$$(2p-1)(p-1) > 0.$$

This holds for $1/3 < p < 1/2$. Hence the desired inequality fails throughout $(1/3, 1/2)$.

Necessity: failure for $1/2 < p < 1$. Take $m = 2$ and $w_1 = w_2 = 1/2$. Let

$$K = v_1 + v_2 \sim \text{Bin}(2, p).$$

For $p > 1/2$,

$$\frac{K}{2} \geq p \iff K = 2.$$

Thus

$$\mathbb{P}\left(\frac{K}{2} \geq p\right) = p^2 < p$$

for every $p \in (1/2, 1)$. Hence the desired inequality fails throughout $(1/2, 1)$.

Combining the sufficiency and necessity parts proves the theorem. □

References

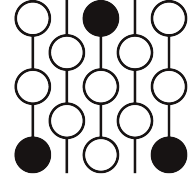
- [1] W. Hoeffding, On the distribution of the number of successes in independent trials, *Annals of Mathematical Statistics* **27** (1956), 713–721.
- [2] K. Jogdeo and S. M. Samuels, Monotone convergence of binomial probabilities and a generalization of Ramanujan’s equation, *Annals of Mathematical Statistics* **39** (1968), 1191–1195.
- [3] I. Pinelis, Optimal binomial, Poisson, and normal left-tail domination for sums of nonnegative random variables, *Electronic Journal of Probability* **21** (2016), article no. 47.

Review of Submission 3C

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. My recommendation is that the submission **should be rejected**. The proof attempt relies on two key inequalities, Lemma 1 and Lemma 2. The main flaw lies precisely in these lemmas: the cited sources either do not contain the claimed statements, or it is not clear how the statements are meant to be derived from them.

Lemma 1 seems especially problematic. Neither of the two cited sources appears to contain a result resembling the claimed lemma. In fact, a quick search suggests that if the strict inequality in the lemma were replaced by a weak inequality, the statement would become a version of Csóka’s conjecture (see <https://arxiv.org/abs/2002.06804>), which, to the best of my knowledge, remains open. This is the major flaw in the proof attempt.

Lemma 2 is also not stated in the cited sources. The solution claims that it is classical and can be proved using Jogdeo and Samuels [2]. This is plausible, but it is not obvious how to obtain the lemma from that paper, and no explanation is provided.

The remaining argument, which reduces the problem to these two lemmas, appears to be correct.

Novelty. I would say that the novelty of this solution is low. Assuming the two lemmas, the reduction is immediate and should be regarded as the main contribution. Since the lemmas themselves already work with the same objects as the target statement, there does not seem to be much new content beyond this reduction.

2 Recommendation

3 Submission with inline comments

Abstract

We determine all $p \in [0, 1]$ such that, for every finite probability vector $(w_1, \dots, w_m)$ and i.i.d. Bernoulli(p) random variables $v_1, \dots, v_m$, one has

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The answer is

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External inequalities used

We use two standard sharp inequalities.

Lemma 1 (Hoeffding–Pinelis comparison for weighted Bernoulli sums). *Let $0 < \theta < 1$, let $\xi_1, \dots, \xi_m$ be i.i.d. Bernoulli(θ), and let $a_i \geq 0$, $\sum_i a_i = 1$. Then for every real x ,*

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where $B_{n,\theta} \sim \text{Bin}(n, \theta)$.

This is the Bernoulli specialization of Hoeffding’s extremal theorem for sums of bounded independent random variables; see Hoeffding [1, Theorem 4]. For the left-tail formulation in essentially the above form, see also Pinelis [3, Corollary 3.7].

Reviewer 2

It does not appear to me that either Hoeffding [1] or Pinelis [3] implies Lemma 1. Moreover, [3] does not contain a Corollary 3.7. In fact, a quick search suggests that Lemma 1 is equivalent to Cs  ka’s conjecture (see <https://arxiv.org/abs/2002.06804v4>). To the best of my knowledge this is still open. This is a major flaw of the attempt.

Lemma 2 (Ramanujan–Jogdeo–Samuels binomial inequality). *If $0 \leq \theta \leq 1/3$ and $B_{n,\theta} \sim \text{Bin}(n, \theta)$, then*

$$\mathbb{P}(B_{n,\theta} < n\theta) \leq 1 - \theta$$

for every integer $n \geq 1$.

Equivalently,

$$\mathbb{P}(B_{n,\theta} \geq n\theta) \geq \theta.$$

This is a classical binomial-tail inequality proved by Jogdeo and Samuels [2]; it is often discussed as a Ramanujan-type binomial inequality.

Reviewer 2

Jogdeo and Samuels [2] prove a stronger result only in the special case where $n\theta$ is an integer. As stated, their paper does not contain Lemma 2. It would not be surprising if the lemma could be derived, with additional work, from [2, Theorem 2.1] via monotonicity; however, no such derivation is provided and it remains unclear what is the source of Lemma 2.

Main theorem

Theorem 1. *Let $p \in [0, 1]$. The following assertion holds for every $m \geq 1$, every choice of nonnegative weights $w_1, \dots, w_m$ with $\sum_i w_i = 1$, and i.i.d. Bernoulli(p) random variables $v_1, \dots, v_m$:*

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This is true if and only if

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For $1/3 < p < 1/2$,

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Therefore

$$\mathbb{P}\left(\frac{K}{3} \geq p\right) = \mathbb{P}(K \geq 2) = 3p^2(1-p) + p^3 = p^2(3-2p).$$

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This holds for $1/3 < p < 1/2$. Hence the desired inequality fails throughout $(1/3, 1/2)$.

Necessity: failure for $1/2 < p < 1$. Take $m = 2$ and $w_1 = w_2 = 1/2$. Let

$$K = v_1 + v_2 \sim \text{Bin}(2, p).$$

For $p > 1/2$,

$$\frac{K}{2} \geq p \iff K = 2.$$

Thus

$$\mathbb{P}\left(\frac{K}{2} \geq p\right) = p^2 < p$$

for every $p \in (1/2, 1)$. Hence the desired inequality fails throughout $(1/2, 1)$.

Combining the sufficiency and necessity parts proves the theorem. □

References

- [1] W. Hoeffding, On the distribution of the number of successes in independent trials, *Annals of Mathematical Statistics* **27** (1956), 713–721.
- [2] K. Jogdeo and S. M. Samuels, Monotone convergence of binomial probabilities and a generalization of Ramanujan’s equation, *Annals of Mathematical Statistics* **39** (1968), 1191–1195.
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